

DSA PROJECT 3

Exact Text Search Using KMP and Rabin-Karp

Abstract

This project implements an exact text searching application using the Knuth-Morris-Pratt (KMP) algorithm, with a separate Rabin-Karp implementation for comparison. Users enter text or upload multiple text files and provide a search pattern. A Java backend performs matching, while an HTML, CSS and JavaScript interface displays the matching positions and search statistics.

KMP constructs a longest proper prefix that is also a suffix (LPS) table and uses it to avoid repeating comparisons after a mismatch. Its matching core runs in $O(N + M)$ time, where N is the total text length and M is the pattern length, with $O(M)$ auxiliary space excluding stored input and results. Rabin-Karp uses rolling hashes to locate candidate matches and verifies candidates character by character. Its expected search cost with few candidates is $O(N + M)$, with $O(NM)$ worst-case time.

The application supports case-sensitive and case-insensitive searches, optional overlapping matches, and match locations with index, line and column. The browser displays highlights, comparison counts and execution time. When a source has no matches, users can edit its text and download an edited copy. A separate KMP console version accepts text or a file path directly in IntelliJ's console. The objective is to demonstrate efficient exact pattern matching through working implementations and visible search results. File loading, position reporting and rendering add overhead beyond the core algorithm.

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